

When RLHF Fails: A Mechanistic Taxonomy of Reward Hacking, Collapse, and Evaluator Gaming

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Abstract

Reinforcement learning from human feedback (RLHF) makes large-scale post-training possible by replacing an underspecified human objective with learned and scalable proxies. The same substitution creates a structured failure surface: optimization can raise the learned reward while external quality falls, degrade both proxy and judge scores, reveal proxy under-alignment, or produce evaluator-specific disagreement. We present an empirical failure-mode study of a compact RLHF pipeline with proximal policy optimization (PPO), direct preference optimization (DPO), uncertainty-penalized PPO (UP-PPO), reward-model uncertainty, approximate policy drift, diversity and repetition diagnostics, and two external LLM judges. Rather than treating reward hacking as a single terminal event, we classify matched transitions between checkpoints using the directions of the learned reward R_ϕ , judge scores $R_1^\dagger$ and $R_2^\dagger$, and their average $\bar{R}^\dagger$. Across 61 checkpoint rows and 1,920 row-level transitions, aggressive PPO has the highest localized reward-hacking rate (14.45%; bootstrap 95% CI: 10.16–18.75%), while UP-PPO yields lower rates in the same aggressive regime (11.33–10.94%). A pre-transition logistic model predicts future row-level reward hacking with ROC-AUC 0.821, and row-level analysis finds localized reward hacking that checkpoint averages miss in 3 of 12 settings. The central conclusion is methodological: RLHF failures are not only final-model pathologies, but training dynamics that can be classified, localized, and partially anticipated.

1 Introduction

RLHF has become a standard mechanism for adapting language models to human preferences, beginning with preference-based reinforcement learning by Christiano et al. [4] and later language-model post-training work by Ziegler et al. [8], Ouyang et al. [12], and Bai et al. [13]. Its effectiveness rests on a consequential approximation: a learned reward model, preference objective, or evaluator stands in for an intended human objective that is only partially observed. This approximation is unavoidable in scalable post-training, but it is also where failures enter. As Amodei et al. [3], Skalse et al. [15], Gao et al. [16], and Pan et al. [19] emphasize in different settings, a policy may exploit the reward model, drift into low-quality regions, over-specialize to an evaluator, or preserve a high proxy score while losing properties that external evaluators or humans would prefer.

The usual phrase “reward hacking” is too coarse for this empirical landscape. A proxy rising while an external evaluator falls is one phenomenon; simultaneous decline of proxy and judge is another; disagreement between two judges is a third. These cases have different signatures and different implications for mitigation. A run that fails because the learned reward is exploitable calls for different evidence than a run whose policy collapses, remains near the supervised reference, or exposes instability in LLM-as-judge evaluation.

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We study RLHF failures as checkpoint-level and prompt-level transitions. Let R_ϕ denote the learned reward-model score, $R^\dagger$ and $R_2^\dagger$ two external judge scores, and $\bar{R}^\dagger = (R^\dagger + R_2^\dagger)/2$ their average. For each evaluated transition, we ask whether the proxy and judges move together or apart. This directional view yields an auditable taxonomy: stable alignment, reward hacking, optimization collapse, proxy under-alignment, conservative stagnation, and evaluator gaming.

The empirical study uses a controlled RLHF pipeline built around GPT-2-scale policies, Anthropic HH-RLHF prompts, PPO variants, DPO, UP-PPO, Monte Carlo dropout uncertainty, approximate KL drift, and two LLM judges. We do not claim universality from a small model. The aim is more precise: to establish a reproducible diagnostic grammar for failures that should be measurable before one argues about scale.

The paper makes four contributions. First, it formalizes a transition-based taxonomy that separates reward hacking from adjacent RLHF failure modes. Second, it shows that row-level diagnostics reveal localized failures hidden by checkpoint averages. Third, it evaluates early-warning models that use only pre-transition signals. Fourth, it compares aggressive PPO with UP-PPO while keeping evaluator disagreement analytically separate from mitigation.

The novelty is not the claim that reward models can be overoptimized; that point is established by Gao et al. and related work. Nor is the contribution a restatement of reward gaming as defined by Skalse et al., feedback-loop reward hacking as studied by Pan et al., or the broad limitations of RLHF surveyed by Casper et al. [21] The contribution is a mechanistic diagnostic layer: a matched-transition taxonomy that distinguishes reward hacking from collapse, proxy under-alignment, stagnation, and evaluator gaming; a row-level localization analysis showing which failures disappear under aggregation; and a two-judge analysis that treats LLM-as-judge disagreement as an empirical failure mode rather than only an evaluation nuisance.

2 Related Work

Learning from preferences. Christiano et al. [4] showed that human comparisons can supervise behavior when scalar ground-truth rewards are unavailable, and Ziegler et al. [8] adapted this preference-learning idea to language models. Stiennon et al. [9], Ouyang et al. [12], and Bai et al. [13] then made instruction following and helpfulness/alignment pipelines central to modern post-training. Bai et al.’s Constitutional AI work [14] and Askell et al.’s alignment-laboratory framing [11] extend the supervision source while preserving the same proxy-optimization structure. Kirk et al. [22] and Lambert et al. [23] further show that post-training choices affect generalization, diversity, and the final behavior of aligned assistants.

Reward hacking and overoptimization. Reward hacking is a modern instance of a much older concern: Goodhart [1] and Campbell [2] argued that optimized measures can stop measuring the intended construct. In AI safety, Amodei et al. [3], Krakovna et al. [10], and Skalse et al. [15] discuss specification gaming and reward misspecification as concrete risks of optimizing learned or hand-designed objectives. Gao et al. [16] show that reward-model overoptimization can follow scaling-like patterns and that larger reward models do not eliminate proxy failure. Our analysis adds a transition-level decomposition: not every undesirable trajectory is reward hacking in the strict directional sense.

Optimization algorithms for alignment. Schulman et al. [5] introduced PPO as a practical policy-gradient update with clipping, and Ziegler et al. [8] and Ouyang et al. [12] helped establish PPO-style updates as a common RLHF optimizer with KL control. Rafailov et al. [17] later proposed DPO, which reframes preference learning as a supervised objective over preferred and rejected completions and avoids an explicit online RL loop. Our experiments compare PPO, DPO, and UP-PPO under the same diagnostic taxonomy.

Table 1: Directional taxonomy of RLHF transition failures. ϵ denotes a small numerical tolerance around zero.

Mode	Directional signature	Interpretation
Stable alignment	$\Delta R_\phi > \epsilon, \Delta R^\dagger > \epsilon$	Proxy and external quality improve together.
Reward hacking	$\Delta R_\phi > \epsilon, \Delta R^\dagger < -\epsilon$	Learned reward improves while judged quality falls.
Optimization collapse	$\Delta R_\phi < -\epsilon, \Delta R^\dagger < -\epsilon$	Optimization degrades both proxy and judged quality.
Proxy under-alignment	$\Delta R_\phi < -\epsilon, \Delta R^\dagger > \epsilon$	Judge improves despite a proxy decline, exposing proxy mismatch.
Conservative stagnation	$ \Delta R_\phi \leq \epsilon, \Delta R^\dagger \leq \epsilon$	Little measurable movement across the transition.
Evaluator gaming	$\text{sign}(\Delta R^\dagger)\text{sign}(\Delta R_2^\dagger) < 0$	The two external judges move in opposite directions.

Uncertainty, monitoring, and calibration. Uncertainty estimates are a natural diagnostic when models operate outside the region supported by training data. Gal and Ghahramani [6] present MC dropout as a simple approximation to Bayesian uncertainty, while Guo et al. [7] study whether model confidence is calibrated to error. In RLHF, uncertainty can be used both as a monitoring signal and as a penalty in the optimized reward, connecting our UP-PPO analysis to broader work on risk-sensitive and conservative optimization. We use MC dropout because it is simple, local to the reward model, and available in the stored artifacts; ensemble and last-layer uncertainty estimators are important alternatives but are not evaluated here.

LLM-as-judge evaluation. LLM judges are increasingly used to evaluate instruction-following systems, but Zheng et al. [18] show that judge-based evaluation brings its own biases, variance, and agreement issues. Dubois et al. [20] further show that length bias can distort automatic preference evaluation. We use those concerns as part of the object of study: two-judge disagreement is not treated only as a nuisance term to average away, but as a diagnosable event. Evaluator gaming occurs when judges move in opposite directions across the same transition, suggesting that measured improvement may be judge-specific. This differs from standard LLM-as-judge benchmarking because the unit of analysis is not a static model ranking, but a training transition in which proxy reward, judge scores, and generated responses move jointly.

3 Failure-Mode Taxonomy

The taxonomy operates on transitions rather than isolated checkpoints. For checkpoint t , let $x_{i,t}$ be the response for prompt i , and let $R_\phi(x_{i,t})$, $R^\dagger(x_{i,t})$, and $R_2^\dagger(x_{i,t})$ be the proxy and judge scores. Aggregate checkpoint metrics average over evaluated prompts; row-level metrics compare the same prompt across two checkpoints. For a transition $t \rightarrow t'$, define

$$\Delta R_\phi = R_{\phi_{t'}} - R_{\phi_t}, \quad \Delta R^\dagger = R_{t'}^\dagger - R_t^\dagger, \quad \Delta \bar{R}^\dagger = \bar{R}_{t'}^\dagger - \bar{R}_t^\dagger. \quad (1)$$

In the implemented taxonomy, the primary judge direction $\Delta R^\dagger$ defines the main class, while $\Delta R_2^\dagger$ identifies evaluator disagreement. Unless otherwise stated, the numerical tolerance is $\epsilon = 10^{-8}$; Appendix E reports sensitivity over larger tolerances. This choice keeps the classifier transparent and avoids hiding judge conflict inside an average.